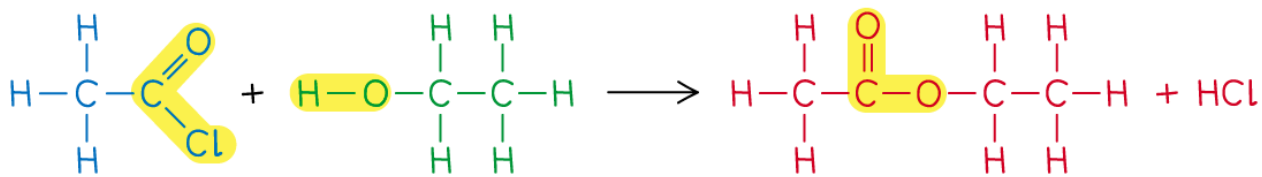


Producing Esters (Cambridge (CIE) A Level Chemistry): Revision Note

Production of Esters by Reacting Alcohols With Acyl Chlorides

- **Esters** are organic compounds with an -COR functional group
- They have **characteristic smells** and are used in perfumes, cosmetics and as solvents
- Esters can be prepared from the **condensation reaction** between alcohols and carboxylic acids
 - This is also called an **esterification reaction**
- A more effective way of preparing esters is from the condensation reaction between alcohols and **acyl chlorides** instead
- Unlike the reactions with carboxylic acids, acyl chlorides are **more reactive** (so the reactions happen **faster**) and their reactions go **to completion** (so no equilibrium mixture is formed and the yield of the ester is maximum)
- Examples of esterification reactions include:
 - Formation of **ethyl ethanoate** from **ethanol** and **ethanoyl chloride**
 - Formation of **phenyl benzoate** from **phenol** and **benzoyl chloride**

Formation of esters from the reaction of alcohols with acyl chlorides



ETHANOYL
CHLORIDE

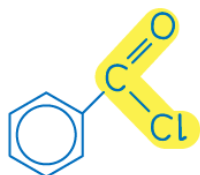
ETHANOL

ETHYL ETHANOATE

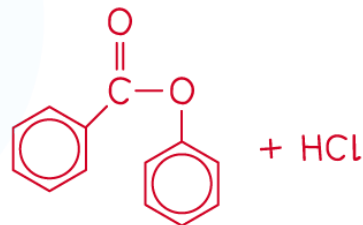
ACYL
CHLORIDE

ALCOHOL

ESTER



+



+ HCl

BENZOYL
CHLORIDE

PHENOL

PHENYL BENZOATE

ACYL
CHLORIDE

ALCOHOL

ESTER

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The first part of the ester name comes from the alcohol and the second part of the ester name comes from the acyl chloride